

EEG-based BMI using Multi-Class Motor Imagery for Bionic Arm

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Abstract— One of the rehabilitation methods developed to help people with physical disabilities, especially patients with disabilities of the upper-limb arm, to do Activities of Daily Living (ADL) is the Bionic arm. The commonly used bionic arm control method is utilizing residual muscle motor neurons. However, the problem arises in patients with upper elbow amputation, limited residual muscle, muscle cells damaged or not growing again which causes the arm to not be able to give enough signals. The electrical activity of the brain recorded using electroencephalography (EEG) can be utilized for bionic arm control. The movement of grasping an object to fulfill ADL is a complex movement, so a system that is able to overcome these needs is needed. The Brain-Machine Interface (BMI) with Motor Imagery (MI) on a bionic arm multiclass control system based on Fast Fourier Transform (FFT) and Support Vector Machine (SVM) is described and discussed in this article. The system built can handle complex movements grasping various objects and evaluated for use with long pauses. The experiment results show good SVM-FFT performance when implemented on a real bionic arm. The use of Additional Movement (AM) increases classification success by up to 20%. On the other hand, this system is able to handle usage even though it has been rested in a few minutes.

Keywords— *Electroencephalograph (EEG), Brain-Machine Interface, Motor Imagery, Support Vector Machine (SVM), Fast Fourier Transform (FFT), Bionic Arm*

I. INTRODUCTION

The World Health Organization (WHO) notes that 15% of the world's population has a disability, between 2.2% (110 million) and 3.8% (190 million) people aged 15 years and older, respectively, experiencing impaired bodily functions/disabilities [1]. The quality of life of people with physical disabilities can decrease due to impaired motion function, one of which is due to amputation [2]. The fact that moving the fingers, either grasping an object or other finger movements is an important and fundamental skill for various ADLs [3], so patients with upper-limb arm disabilities require special treatment to handle this problem. Special treatment for patients with disabilities can be done with rehabilitation medical equipment in the form of prosthetic. Patients with upper limb defects usually use prosthetic arms that have a variety of models, but technological advancements allow the demand and application of prosthetic bionic arms (bionic arms) to continue to increase. So that the development of the bionic arm also continues to increase both in terms of capability and reliability, as well as the materials used.

There are several sensor-related approaches that are used as bionic arm controls. Myoelectric control works by changing the contact of motor neurons with muscle fibers in neuromuscular. Until now, myoelectric has become the most widely used control method in the bionic arm [4]. Signals measurement using this method is easily done by calculating muscle strength, but this method cannot always be installed

for all people with disabilities or hand amputations. In some cases such as amputation of the elbow, patients with limited residual muscle, muscle cells die or cannot grow again causing the arm to not provide enough of signals to control the prosthetic arm [5]. Another weakness of this method is its sensitivity to interference that limits its operational scope [6].

Another alternative is EEG-based BMI. This system measures the electrical activity of the brain as a signal to control something, so it does not depend on muscles to modulate bionic arm control. [7]. The modulation of brain signals to move an object without having to move a limb is called Motor Imagery (MI) [8]. EEG is a non-invasive technique, so as long as the patient has normal brain function, this technique can be utilized as a means of BMI [9]. The BMI approach has been utilized in various fields, for example, controlling a robot vehicle even for rehabilitation such as wheelchairs and bionic arm [10].

In its development, EEG-based BMI-MI has various challenges, one of which is control instructions to facilitate users in operating the device. The MI usually use by user intention as a device control instruction, but this method is draining so that the user feels tired quickly. Therefore, one way to simplify control instructions is use Additional Movement (AM).

Because of the benefits offered by EEG-based BMI-MI, many studies offer various methods with their advantages and disadvantages. Bionic arm control using conventional EEG sensors offered by Hayashi and Kiguchi [11] achieves a success rate of up to 80%. The use of conventional EEG sensors is also used by several other researchers for an in-depth analysis of waves generated from each lobe in MI [12]. However, the use of conventional EEG sensors connected by cables to the data processor causes discomfort and reduces user mobility in meeting ADL.

In addition to adequate portability to meet ADL, portable EEG sensors [13] also show results that are not inferior to conventional EEG sensors. The results obtained are also determined because of the right handling in processing the required data. Zen li et al [14] used Muse 2 Headband as a portable EEG sensor by utilizing gamma waves from the AF7 and AF8 channels. SVM is used as a classification method. The output obtained to drive the 2-class imagery motor pattern with a percentage of 95.1%. For complex ADLs, however, 2-class as a representation of 2 movements is not able to fulfill it as a whole. Therefore more pattern classification is needed.

Riyadi et al [15] proposed a 5-class imagery motor pattern using the same method. The results obtained show that the proposed method can function well to recognize more patterns. However, this research is limited only as a simulation. However, for the system to run properly, several parameters need to be considered. In this study, the real

implementation of bionic arm control is proposed by using AM as control instructions. The proposed system prioritizes portability and user comfort for ease of meeting ADL, but with good system success.

II. RESEARCH METHODS

A. Subject

In this study, subjects had previously been confirmed to be healthy and had practiced several times to test the system. The sensor used is Muse Headband which has 4 main channels and 1 reference channel. The EEG sensor used is a dry electrode with wireless communication so it is easy to maintain and use to meet ADL because it is not interrupted with multiple cables. The electrode placement is based on the International System 10-20 shown in Figure 1. The position of the Muse Headband electrodes is on the AF7, AF8, TP9 and TP10 channels based on the average adult head size, and FPZ as the reference electrode position. Furthermore, AF7 and AF8 electrodes are located in the Frontal lobe which has one of functions related to movement. While the TP9 and TP10 electrodes located in the Temporal lobe are associated with perception and recognition [16], [17]. These lobes have the potential to produce strong signals for MI purpose.

B. Data Acquisition And Preprocessing

In measuring the neurophysiological rhythmic activity of MI-based EEG signals, the most dominant modulation occurs in the sensorimotor cortex. The rhythmic activity that occurs in the sensorimotor cortex is referred to as sensorimotor rhythms (SMRs). With SMR, MI can be coded to form the nerve control used in BMI [18]. The use of an EEG sensor with electrodes located on the sensorimotor as MI control can reduce computational time because it only uses the signals needed, rather than using an EEG sensor with electrodes located in the cortex which produces a small signal potential for MI.

EEG signals basically consist of a number of frequency waves that appear under different conditions. Gamma waves (30-44 Hz) can occur due to increased feelings of happiness and are related to cognitive or motor function, Beta waves (13-30 Hz) appear when a person feels nervous or scared and is also caused due to mental activity, Alpha waves (8-13 Hz) is emitted when a person feels mentally and physically relaxed, Theta waves (4-8 Hz) occur when someone is unconscious, while the Delta wave (1-4 Hz) occurs when in a state of deep sleep and meditation [19].

To ease the computational burden, it is only necessary to use signals related to motor movements. Therefore, in this study, the Gamma signal was chosen as the input signal. A block diagram showing the stages of the whole system from signal acquisition to bionic arm control is shown in Figure 2.

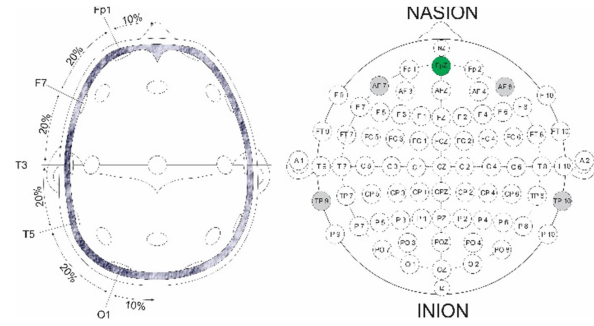


Fig. 1. Electrode placement is based on International System 10-20.

Brain waves are captured by Muse EEG 2 Headband then sent to Raspberry Pi via Bluetooth. The raw data obtained is calculated to get the absolute features and power extraction of brainwave frequency bands from Gamma waves. Then, the extracted features are sent to the classification subsystem for control commands. The characteristics received by Raspberry Pi are processed based on Python scripts. Just like the previous research, in this study, Gamma waves are used in the system because they provide better results with higher accuracy.

The bionic arm is made with a 3-D printing technique with linear actuator as the main mover. Equipped with ATMEGA as the main microcontroller and Lipo Battery for power supply. Communication between the bionic arm and the Raspberry Pi uses Bluetooth via a sub-controller. Raspberry Pi as the main controller processes brain wave signals and classifies, while the sub-controller functions for the strategy of sending movement commands. This is done so that the performance of the Raspberry Pi classification is not burdened with other work so that it is more optimal.

C. Feature Extraction

The success rate of the system depends on the performance of the classification method and feature extraction used [20]. Both methods are correlated to get maximum results. In BMI systems, feature extraction is an important part of signal processing. Characteristic is a form of new data from raw data after going through the extraction process that represents the measurement attributes of the signal [21]. Characteristics of BMI have noise, high-dimensional vectors and contain time information because the pattern of brain activity is related to the time domain. So the BMI characteristic is non-stationary because it changes with time [22]. The initial process of feature extraction is by recording raw data.

EEG data is taken with a 256 Hz sampling frequency. Then the signal discontinuity is minimized by the Hamming Windowing process at the beginning and end of each frame by

$$w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), 0 \leq n \leq N-1 \quad (1)$$

$$y(n) = y \times w(n) \quad (2)$$

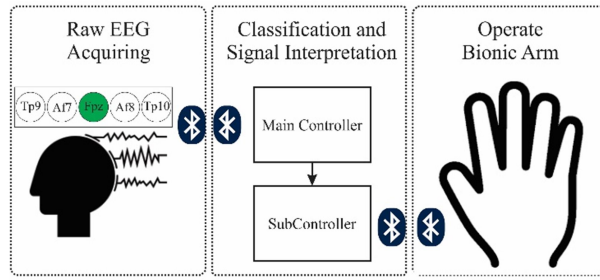


Fig. 2. Block diagram of the system.

After the raw data is obtained, feature extraction is then performed. Feature extraction is done to minimize the loss of important information contained in the signal and find a good representation of the EEG signal, as well as maintain the most relevant properties according to the brain activity carried out [20], [23].

Because the EEG signal is captured on a time basis, it needs to be converted to a frequency base using FFT before extracting important information from the signal. FFT has fast computing capabilities that work by converting signals from the time domain to the frequency domain [24]. Different mental activities produce different powers in different frequency ranges and are displayed by Power Spectral Density (PSD) [25]. PSD shows energy power as a function of frequency. The energy in several frequency ranges can be obtained by integrating PSD with that frequency range [26]. PSD which has been calculated will produce coefficient values. This coefficient value is used as input to the classification method to determine what movement will be performed. The characteristics of the EEG signal obtained are calculated by estimating the PSD as an EEG signal sample [23]. PSD is calculated by Fourier who transforms the autocorrelation order estimation using the Welch method as a non-parametric method. The order of information $x_i(n)$ is stated as

$$x_i(n) = x(n + iD), \quad n = 0, 1, 2, \dots, M-1 \quad (3)$$

where $i = 0, 1, 2, \dots, L-1$

the iD is used as the starting point of sequence i . Then L with a length of $2M$ represents the data segment formed. The resulting periodogram gives by

$$P_{xx}^{(i)}(f) = \frac{1}{MU} \left| \sum_{n=0}^{M-1} x_i(n) w(n) e^{-j2\pi f n} \right|^2 \quad (4)$$

In this window function, U gives the power normalization factor and is chosen as such

$$U = \frac{1}{M} \sum_{n=0}^{M-1} w^2(n) \quad (5)$$

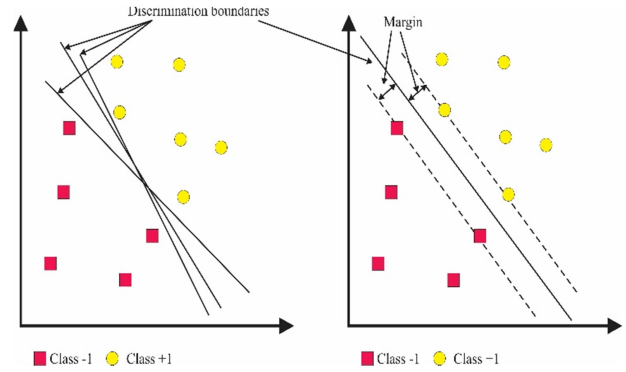


Fig. 3. SVM finds the best hyperplane.

where $w(n)$ is a window function. This modified average periodogram gives the Welch power spectrum

$$P_{xx}^W = \frac{1}{L} \sum_{i=0}^{L-1} P_{xx}^{(i)}(f) \quad (6)$$

D. Classification Method

SVM aims to find a separating hyperplane by maximizing margins to increase generalization, i.e. the distance of the closest training point as shown in Figure 3. The basic concept of the SVM algorithm is to map input space to feature space to separate linearly [27], [28]. The hyperplane is looking for an optimal hyperplane that separates two classes in the input space. The optimal hyperplane is a combination of support vectors from the optimal hyperplane or several feature points.

Learning in classification aims to find the line (hyperplane) that separates between classes. As shown in Figure 3, the left side has several alternative discrimination boundaries. While Figure 3 shows the right side hyperplane located in the middle of the two classes which are the best hyperplane. Hyperplane margins are used to measure the best Hyperplane. While the margin is the distance between the hyperplane and the nearest data for each class.

The SVM algorithm shows good results in various experiments in both linear and non-linear forms, in binary and multiclass BMI with only a simple training set. SVM regularization is able to handle the noise and outliers that arise and can improve the ability to generalize classification. Another advantage of SVM is simple linear functions that have high stability. The stability of this SVM is able to handle the instability of BMI characteristics which can complicate classification.

SVM based on statistical learning theory can build optimal separating hyperplane in high dimensional space to classify new samples. Training data sets are $\{(x_1, y_1), (x_2, y_2), \dots, (x_l, y_l)\}$ where $x_i \in \mathcal{R}^n$ shows a sample with n -dimension, and $y_i \in \{+1, -1\}$ is a class sample x_i . The training sample is transformed into a higher dimensional feature space by the mapping function so that it finds the optimal hyperplane. The separating hyperplane that is represented by

$$w \cdot \phi(x) + b = 0 \quad (7)$$

The SVM decision function is represented as

$$f(x) = \text{sign} \left(\sum_{i=1}^l y_i K(x_i, y_j) + b \right) \quad (8)$$

where l is the number of support vectors, and $K(x_i, y_j)$ is the kernel function.

SVM performance shows good results for classifying two classes. In applications with non-linear input data that require classification of more than two classes, a kernel trick is used to improve SVM performance [29]. Kernel trick is used to map the initial dimensions (lower dimensions) of data sets to new dimensions (relatively higher dimensions). In this research, the Radial Basis Function (RBF) is used as the SVM algorithm kernel, ie

$$K(x_i, y_j) = \exp \left(\frac{\|x_i - x_j\|^2}{2\sigma^2} \right) \quad (9)$$

III. RESULTS AND DISCUSSION

This research is intended to move the bionic arm from five classifications of motor imagery results. Figure 4(a) shows the system training and testing process carried out by a subject. The experiment began with a classifier training session by giving five different patterns of brain activity to motor imagery. In the training session, the subject will distinguish five different patterns for bionic arm control commands. Each pattern is given AM to facilitate the user in its operation. The training session was conducted for 2 minutes 30 seconds, consisting of the first 10 seconds of preparation, 20 seconds of a recording of each pattern and 10 seconds of pauses between patterns. The device is equipped with a sound buzzer to give alert the user when starting a training session, at each recording pattern, and at the end of a training session.

In both training and testing sessions, each subject is done by sitting relaxed and watching the monitor screen that will display a direction icon to help the intention in the training stage. The percentage of training success will appear after the training phase is complete. The next stage is to control the bionic arm with the only intention. This method will then be compared by including AM as a control instruction.

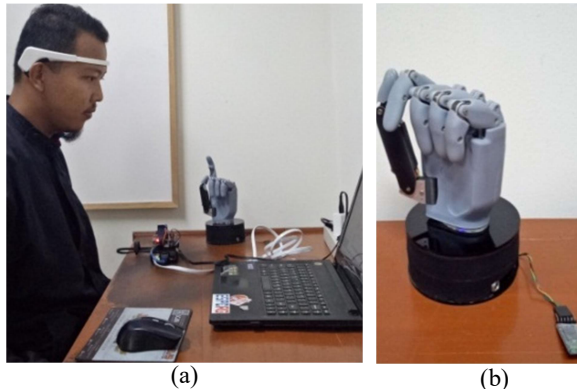


Fig. 4. Training and testing process (a) the subject operates the system and (b) the position of the bionic arm when "Key-Grip" movement.

TABLE I. BIONIC ARM MOVEMENT BASED ON CLASS

Class	Bionic Arm Movements	Functions
Class 1	Release	Initial position and to release the object
Class 2	Key-Grip	To hold the key
Class 3	Finger Point	To press the switch or button
Class 4	Precision Open	To open something small
Class 5	Spoon-Hold	To hold the spoon

TABLE II. COMPARISON OF SYSTEM TEST RESULTS

Test	Without AM		With AM	
	Prediction (%)	Actual (%)	Prediction (%)	Actual (%)
1	78.57	80	91.76	90
2	70.52	70	95.24	100
3	72.62	70	90.59	90
4	78.84	80	94.05	100
5	86.00	90	96.04	100
Average	77.31	78	93.54	96

Five different classes of bionic arm movements resulting from the test are shown in Table I. Every bionic arm movement is a basic movement to meet the ADL. This study shows a satisfying success rate using the FFT-SVM method which is implemented on a real bionic arm. All bionic arm movements used are based on the daily movements of normal activities. Hand movements begin with a "Release" motion. The next movement depends on the user's command as needed. However, to change from one movement to another movement (besides the "Release" movement), for example, the "Key-Grip" to "Finger Point" movement, must go through the "Release" movement command first. This is used so that the fingers of the bionic arm do not collide with each other when the change between movements. On the other hand, this stage will also make it easier for users to position themselves against the object. Furthermore, by avoiding collisions between the fingers of the bionic arm, it can secure the device from damage such as cracks in the frame, broken or bent from the connecting plate between joints or broken on the actuators.

The imaging and prediction classes processed by Raspberry Pi are displayed on the monitor screen and as a bionic arm control output. Table II shows the results of the tests carried out by each subject without AM and with AM. The subject was tested five times. Prediction results appear in each test and will be displayed on the monitor screen after the training process is complete. For actual results, the test is carried out by moving the bionic arm ten times to all movements. The actual test on the bionic arm is performed on each test without AM and with AM.

The tests show the classification prediction with AM gives better results than without AM. Testing with AM significantly increases the success of the classification on predictions and actual. The actual grasping movement test is also carried out to verify the classification prediction and show the results are not much different from the predictions obtained. This test is done by moving the bionic arm to all possible movements of 10 times. Correlation testing is done by giving 5 minutes pause between tests to ensure the system can run well even if it is interrupted by other activities that do not involve the device. Correlation testing for 5 times in a

row at a prediction of 94.05% resulted in an actual success of 100%. This shows that the results of the classification of the system that was built were not interrupted despite being given a pause of use for the time as was tested.

Overall system results show good performance. Compared to the results of studies using EEG Muse Headband and SVM as the same EEG sensors and classification method, researchers [14] were able to provide a classification success rate of 95.1% in the 2-class imagery motor pattern. Whereas in this study, the 5-class imagery motor pattern was able to produce an average classification success of 93.54%. Considering the complexity of classification with more classes, the results of this system are not too far behind. Even in the actual testing to move the bionic arm, this system was able to obtain a success rate of 96%. Similar with the results obtained by researchers [15] with 5-class imagery motor patterns obtained through simulation. However, in this study, the 5-class imagery motor pattern was significantly applied to the bionic arm. So that the result of system tests that have been carried out can be verified and proven significantly by bionic arm movements.

IV. CONCLUSION

Based on the results of tests that have been carried out, it can be concluded that the brainwave classification system that is assisted by AM as a command to control the bionic arm into five classes, shows good results. The SVM classifier is a classification algorithm with the RBF kernel is able to classify properly. In this study, it was found that AM added as a support for control commands can improve the classification results for the better. In the case of this assessment, AM increases the classification prediction by up to 20%. The addition of AM will also make it easier for bionic arm users to operate it and reduce fatigue during training and use. Correlation testing with time is also carried out to test the system if the use of equipment is subjected to a long pause but at one training only. The results obtained are the system is able to handle these problems properly.

Although the system has been validated by testing by providing a long time lag in one training and getting satisfactory results, this study can be further developed by conducting ADL testing directly by patients with disabilities. Testing is done in a few days or weeks in order to get convincing validation. This will provide an opportunity to develop a robust bionic arm that is controlled based on MI with portable EEG.

Furthermore, this research can be developed to get a robust system. That is, to produce brain wave-based bionic arm with only one subject to conduct initial training and can be used by several user without doing training again. This will increase the effectiveness in terms of user time.

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